Co-evolution of robots

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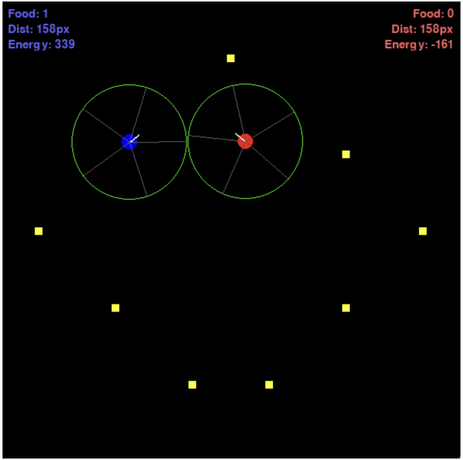
February 17, 2026



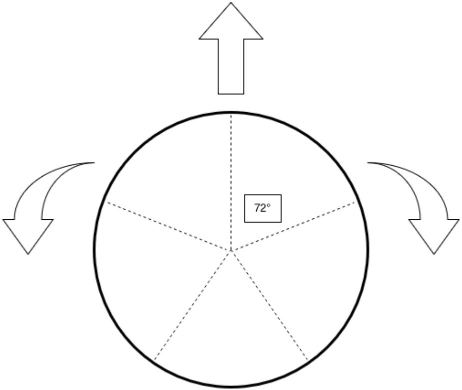
A **Robot Duel** is a game where two controlled robots fight in an arena under certain rules and conditions. In this scenario:

- The robots are controlled by neural networks;
- Robots have a default energy, which is reduced by actions and attacks;
- Energy points give boost of energy;
- If robots collide the one with higher energy wins;
- There is a time limit.

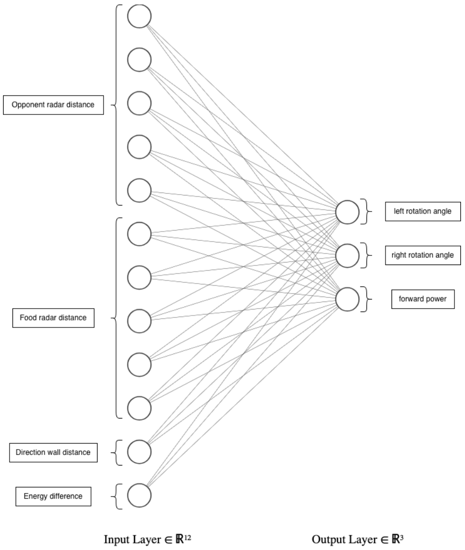
The robot

The robot has 5 sensors that track information regarding the position of the opponent and the nearest energy points. The sensors are split in the 2π and each is responsible for $\frac{2\pi}{5}$ degrees of the space.
To move the robot, three actions are available:

- rotate left;
- rotate right;
- move forward.

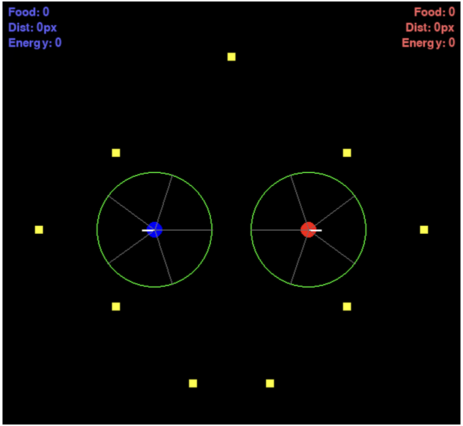


Robot controller



For each state **s** of the game, the robot controller has to choose an action **a** to perform. The state is composed of the radar information, the distance to the wall the robot is facing and the energy difference with the opponent.

The duel



- 600x600 arena;
- 7 fixed energy points;
- no default energy;
- 500 boost of energy for each energy point;
- energy loss depends on forward distance and rotation angle;
- 750 time steps limit.

NEAT

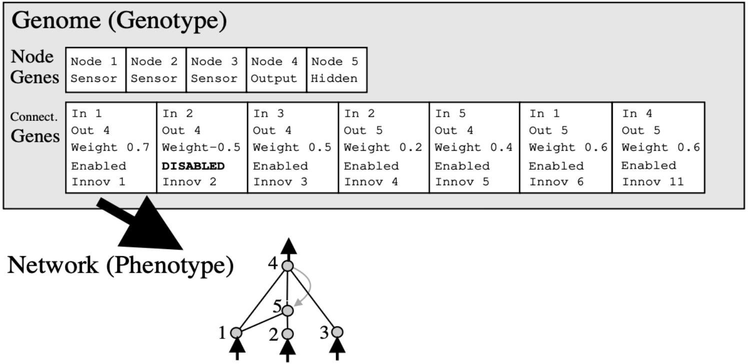
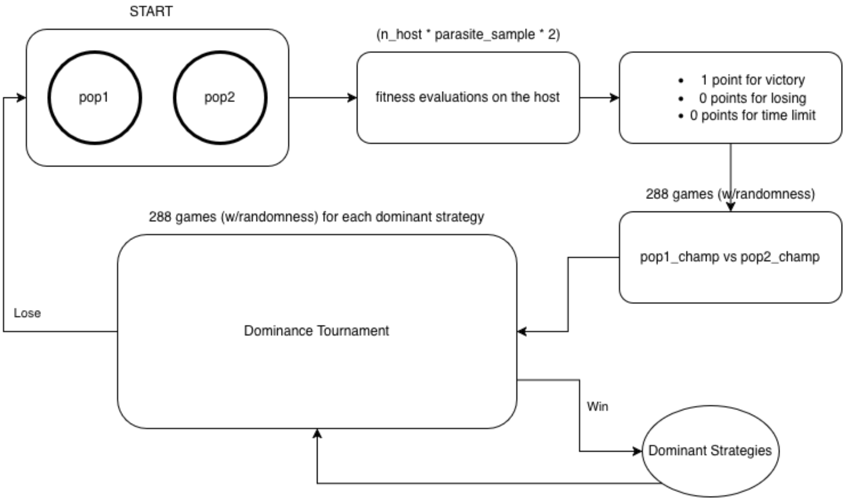


Figure 1: A NEAT genotype to phenotype mapping example [Stanley and Miikkulainen, 2004].

NEAT config

```
[NEAT]
fitness_criterion      = max
fitness_threshold      = 50
pop_size               = 512

[DefaultGenome]
activation_default      = sigmoid

conn_add_prob           = 0.1
conn_delete_prob        = 0.0

feed_forward            = False
initial_connection      = full

num_hidden              = 0
num_inputs              = 12
num_outputs             = 3

...
```

Co-evolution setup



Training

I trained the system for 100 generations locally on my machine (Apple M3, 16GB RAM), taking around 7.5 hours to complete.

- 2 populations;
- 512 individuals each;
- parasite sample size of 12 (4 highest species champions, 8 random hall of fame of all generations' champions);
- 144 games for each starting position to determine the generation champion and for the dominance tournament;
- 2 extra energies for each of the 144 games for robustness.

```
Gen 13 | ... | Total: 273.2s
Generation champion beats dominant strategy #1 with score 288 vs 0
Generation champion beats dominant strategy #2 with score 146 vs 120
Generation champion beats dominant strategy #3 with score 147 vs 131
Generation champion beats dominant strategy #4 with score 277 vs 11
*** NEW DOMINANT STRATEGY #5 ***
```

```
Gen 75 | ... | Total: 282.2s
Generation champion beats dominant strategy #1 with score 288 vs 0
Generation champion beats dominant strategy #2 with score 50 vs 0
Generation champion beats dominant strategy #3 with score 223 vs 24
Generation champion beats dominant strategy #4 with score 220 vs 0
Generation champion beats dominant strategy #5 with score 62 vs 0
*** NEW DOMINANT STRATEGY #6 ***
```

Demo time!

Conclusion and future work

This project demonstrates the potential of co-evolutionary algorithms in developing competitive strategies for robot duels:

- NEAT effectively evolves neural network controllers for the robots;
- Co-evolution promotes the emergence of increasingly sophisticated strategies as each population adapts to the other;
- Future work could explore more complex robot designs, additional sensors, and more fine-tuning of the NEAT parameters to further enhance performance;
- Further training could be performed to exploit different strategies discovered by NEAT.

Thank you for listening !

References I

[Stanley and Miikkulainen, 2004] Stanley, K. O. and Miikkulainen, R. (2004).
Competitive coevolution through evolutionary complexification.
Journal of artificial intelligence research, 21:63–100.